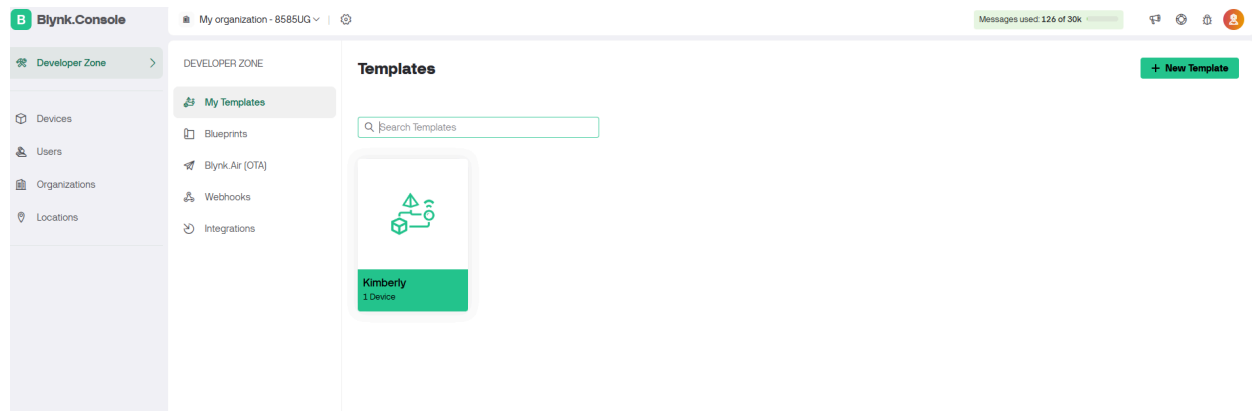


TUTORIAL COMPLETION(1-6):

3.



NAME

Kimberly

8 / 50

HARDWARE

Arduino

CONNECTION TYPE

WiFi

TEMPLATE DESCRIPTION (OPTIONAL)

This is my template

19 / 128

Devices

New Device

Choose a way to create new device

From template



Scan QR code



Manual entry



 Point on the cards to see instructions

Cancel

New Device

Create new device by filling in the form below

TEMPLATE

Kimberly

DEVICE NAME

Kimberly 8 / 50

Cancel Create



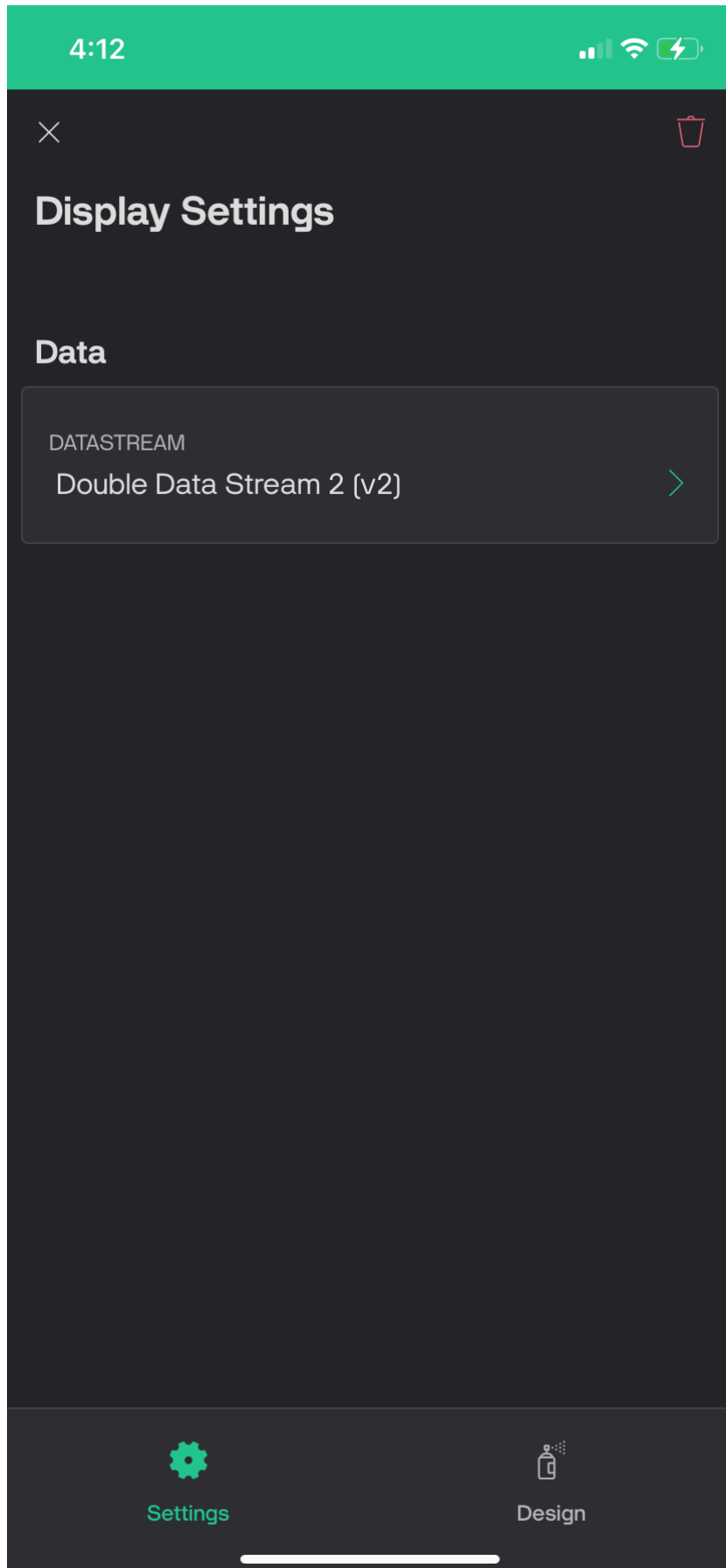
Kimberly

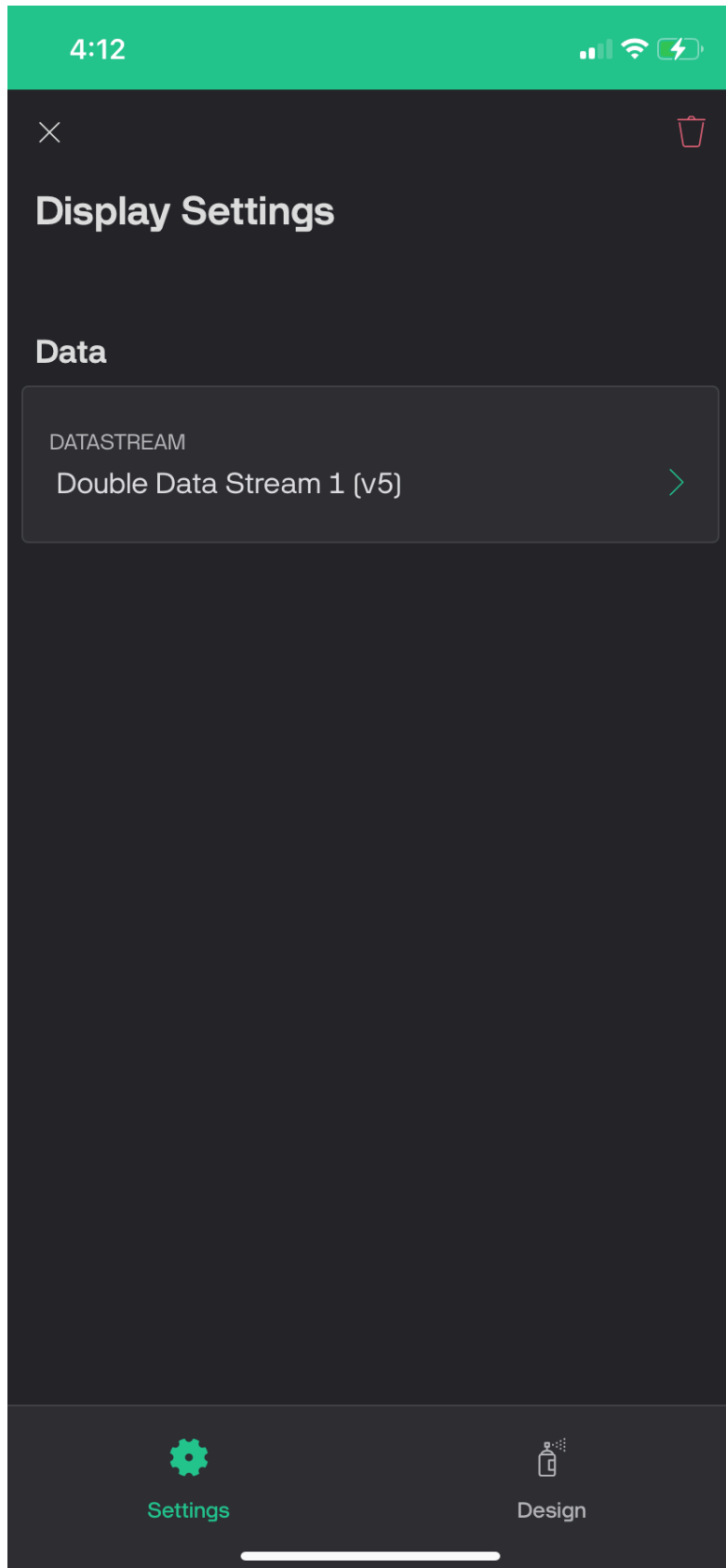
... Edit

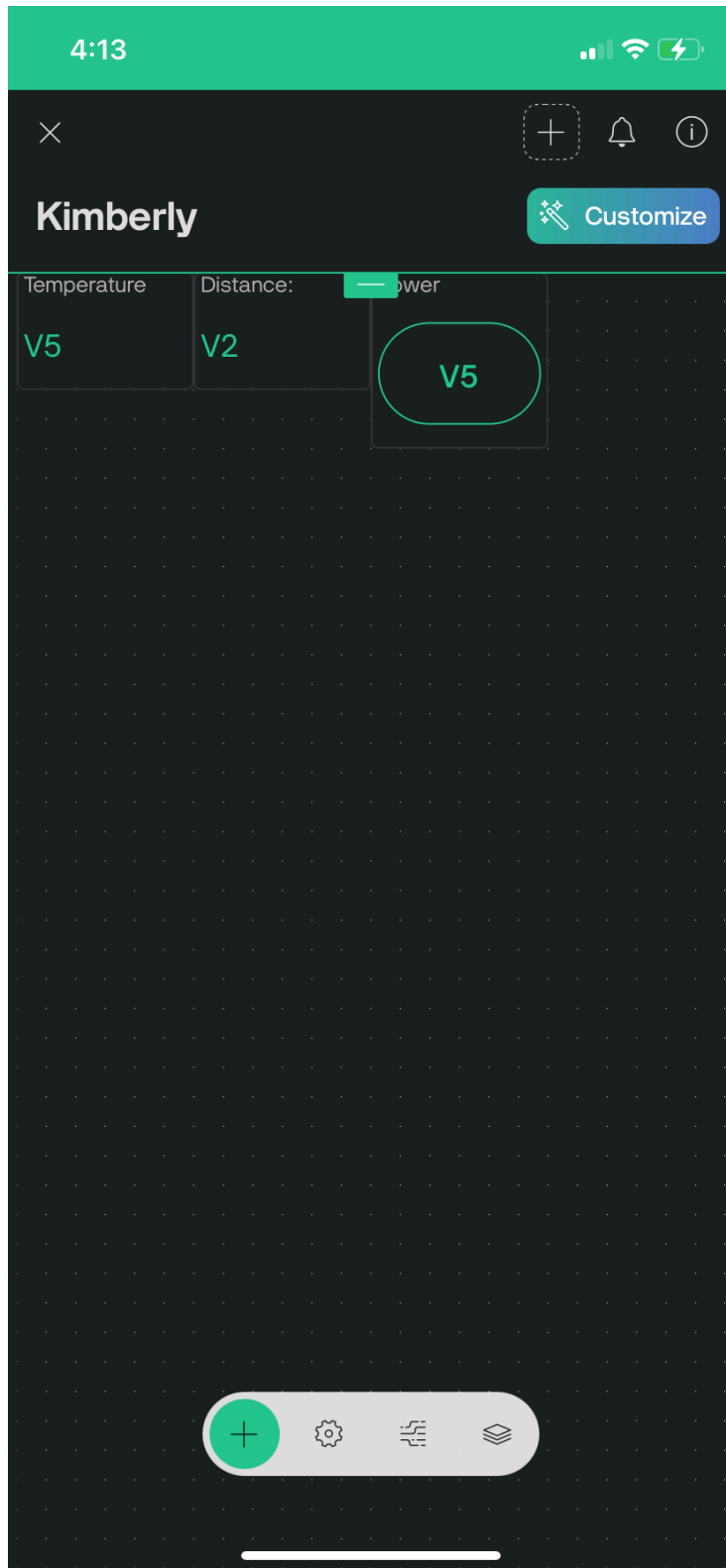
Datastreams

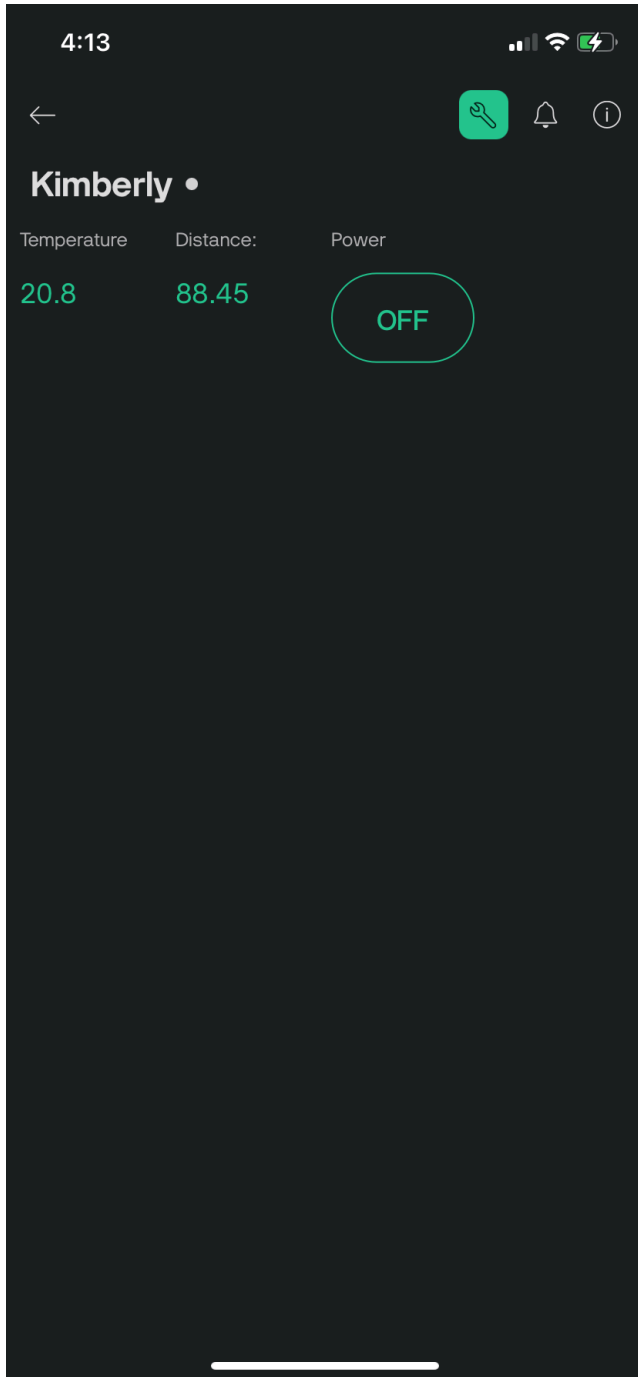
Search datastream

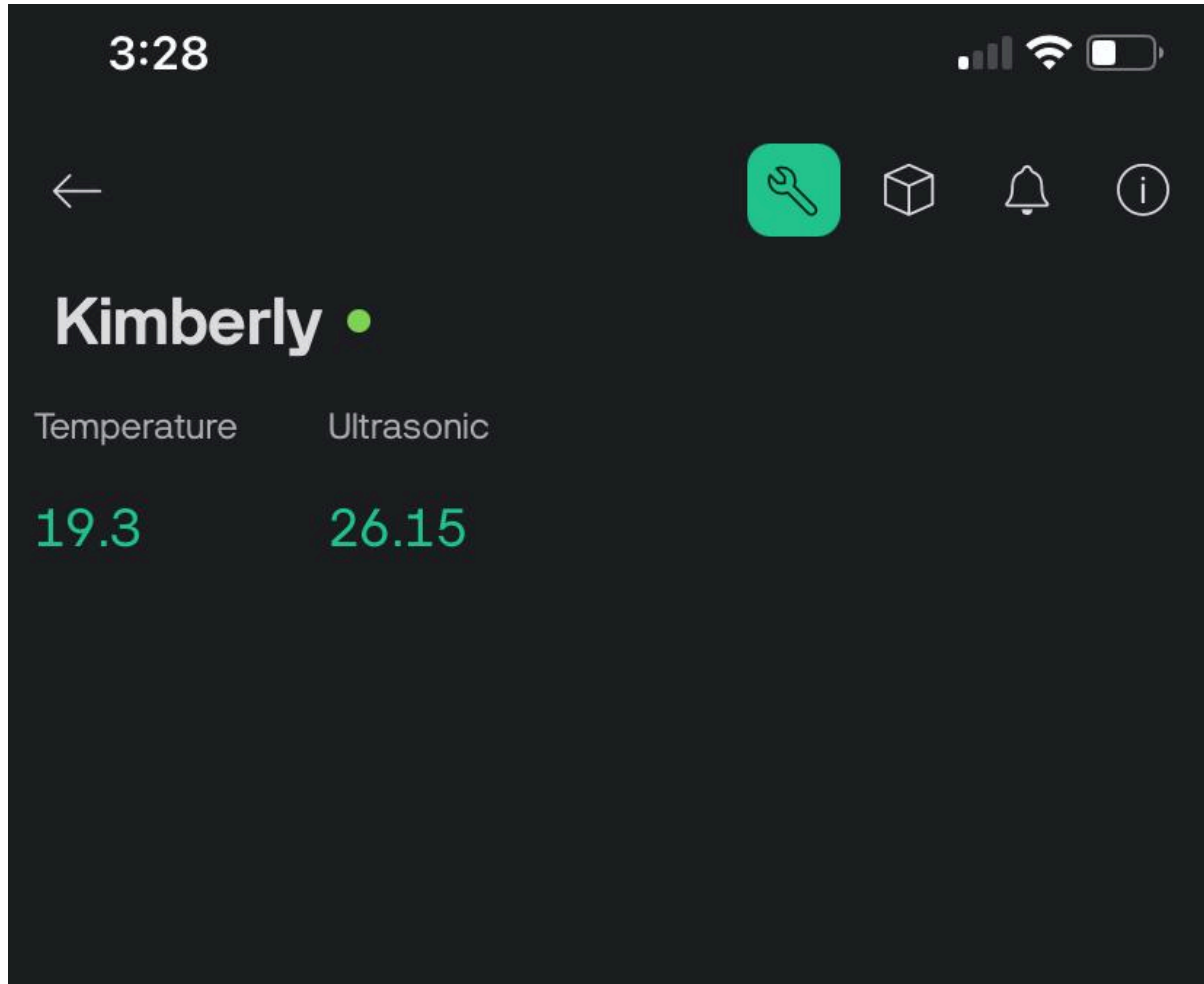
Id	Name	Alias	Color	Pin	Data Type	Units	Is Raw	Min	Max	Decimals	Default
1	Double Data Stream 1	Double Data Stream 1		V5	Double		false	-20	125	###	
2	Double Data Stream 2	Double Data Stream 2		V2	Double		false	0	100	###	











4.

SENSORS CODE:

```
// Template ID, Device Name and Auth Token are provided by the Blynk.Cloud
// See the Device Info tab, or Template settings
#define BLYNK_TEMPLATE_ID "TMPL2KIo5FwVI"
#define BLYNK_TEMPLATE_NAME "Kimberly"
#define BLYNK_AUTH_TOKEN "9hP9yGXO3NndpLtPTAjbZ_Ekuo-yn7BH"
#define tempPin A1
const int trigPin = 6;
const int echoPin = 7;
long duration = 0;
float distance = 0.0;
```



```
// Comment this out to disable prints and save space
#define BLYNK_PRINT SerialUSB
#include <SPI.h>
#include <WiFi101.h>
#include <BlynkSimpleWiFiShield101.h>
// Your WiFi credentials.
// Set password to "" for open networks.
char ssid[] = "RojasFam";
char pass[] = "L0v3sh0t";
char auth[] = "9hP9yGXO3NndpLtPTAjbZ_Ekua-yn7BH";

void setup()
{
    // Debug console
    SerialUSB.begin(115200);
    Blynk.begin(auth, ssid, pass);
    // You can also specify server:
    //Blynk.begin(auth, ssid, pass, "blynk.cloud", 80);
    //Blynk.begin(auth, ssid, pass, IPAddress(192,168,1,100), 8080);
    pinMode(tempPin, OUTPUT);
    pinMode(trigPin, OUTPUT);
    pinMode(echoPin, INPUT);
}

void loop()
{
    Blynk.run();
    sensor();
}

void sensor() {

    //temperature calculation
    float heat = analogRead(tempPin) ;
    float v = heat * (3300/1024);
    float realHeat = (v - 500 ) / 10;

    //acquire distance
    digitalWrite(trigPin, LOW);
    delay(1);
```

```
digitalWrite(trigPin, HIGH);
delay(1);
digitalWrite(trigPin, LOW);

//distance calculation
duration = pulseIn(echoPin, HIGH);
distance = (duration * .034) / 2;

//prints the temperature
Serial.print("temperature : ");
Serial.print(realHeat);
Serial.println(" C \n");
Blynk.virtualWrite(V5, realHeat); //collect the value and bring it to v5

//prints the distance
Serial.print("Distance: ");
Serial.print(distance);
Serial.println(" cm");
Blynk.virtualWrite(V2, distance);
delay(2000);
}
```

TEMPERATURE BUTTON CODE:

```
// Template ID, Device Name and Auth Token are provided by the Blynk.Cloud
// See the Device Info tab, or Template settings
#define BLYNK_TEMPLATE_ID "TMPL2KIo5FwVI"
#define BLYNK_TEMPLATE_NAME "Kimberly"
#define BLYNK_AUTH_TOKEN "9hP9yGXO3NndpLtPTAjbZ_Ekuo-yn7BH"
#define tempPin A1
// Comment this out to disable prints and save space
#define BLYNK_PRINT SerialUSB
#include <SPI.h>
#include <WiFi101.h>
#include <BlynkSimpleWiFiShield101.h>
// Your WiFi credentials.
// Set password to "" for open networks.
char ssid[] = "RojasFam";
char pass[] = "L0v3sh0t";
char auth[] = "9hP9yGXO3NndpLtPTAjbZ_Ekuo-yn7BH";
```

```
void setup()
{
    // Debug console
    SerialUSB.begin(115200);
    Blynk.begin(auth, ssid, pass);
    // You can also specify server:
    //Blynk.begin(auth, ssid, pass, "blynk.cloud", 80);
    //Blynk.begin(auth, ssid, pass, IPAddress(192,168,1,100), 8080);
    pinMode(tempPin, OUTPUT);
}

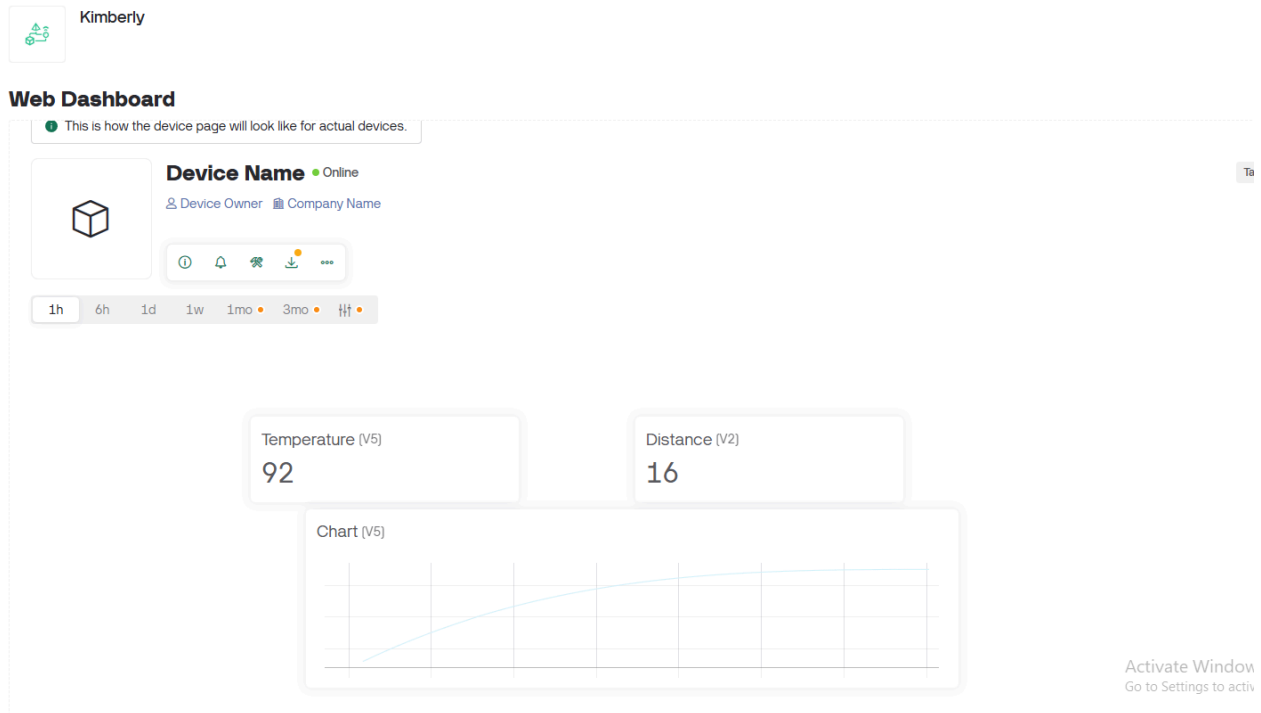
BLYNK_WRITE(V0)
{ // Executes when the value of v1 changes remember screen above
    if(param.asInt() == 1)
    { //if it is on
        sensor(); //see the value of the sensor
    }
}

void loop()
{
    Blynk.run();
}

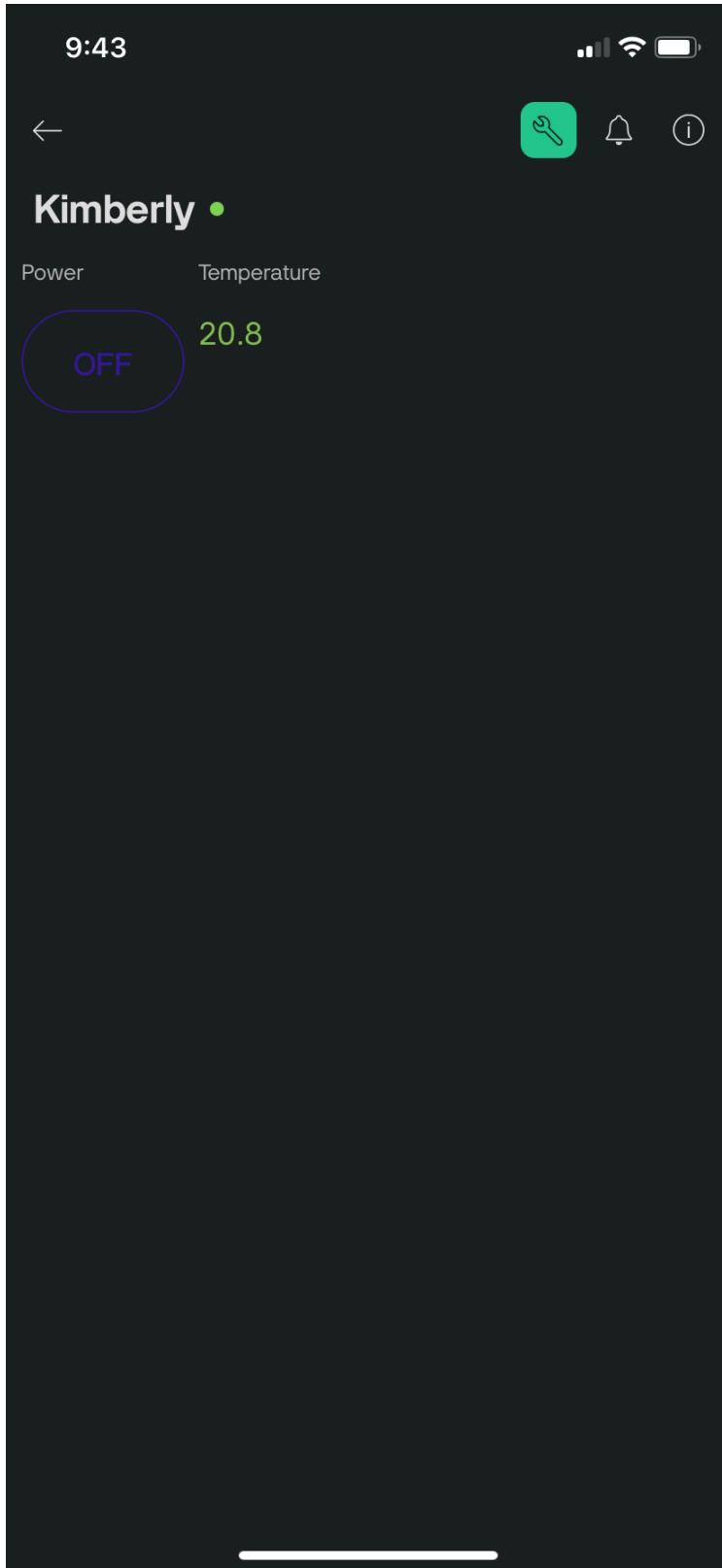
void sensor()
{
    //TEMPERATURE CALCULATION
    float heat = analogRead(tempPin) ;
    float v = heat * (3300/1024);
    float realHeat =(v - 500 ) / 10;

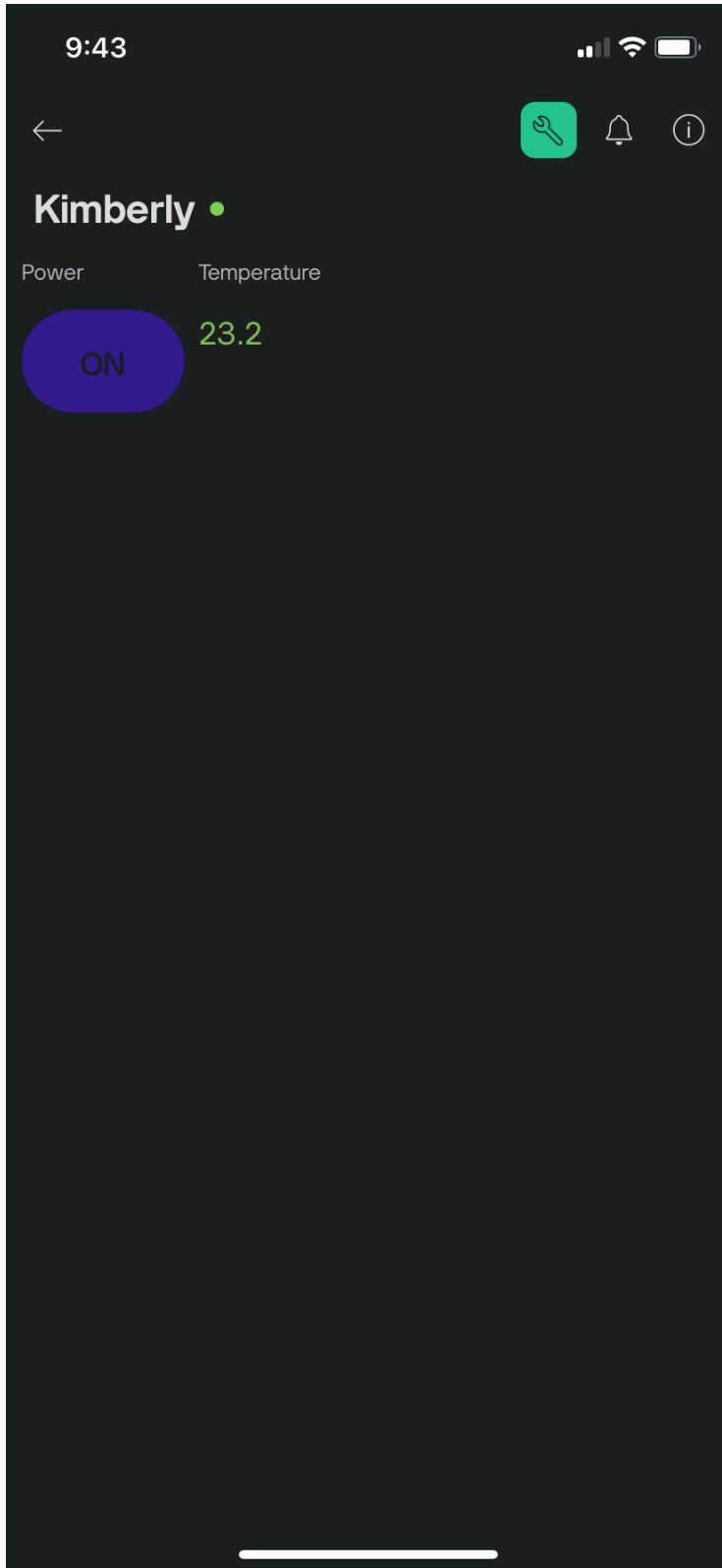
    //TEMPERATURE PRINT
    Serial.print("temperature : ");
    Serial.print(realHeat);
    Serial.print(" C \n");
    Blynk.virtualWrite(V1, realHeat); //collect the value and bring it to v5
    delay (2000);
}
```

5.



6.

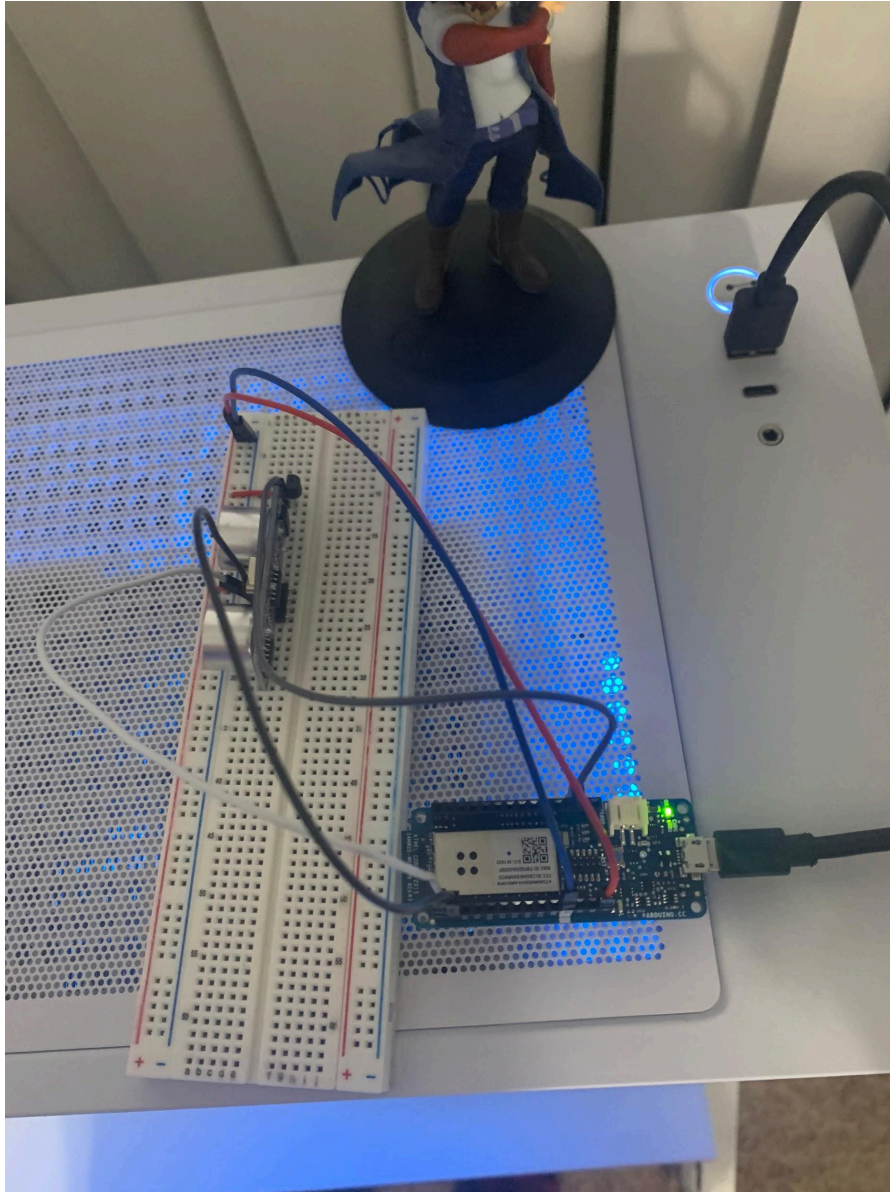






```
21:09:44.115 -> [739] Connecting to blynk.cloud:80
21:09:45.545 -> [2202] Ready (ping: 88ms).
21:10:02.417 -> [1692] Ready (ping: 89ms).
21:41:37.463 -> temperature : 20.80 C
21:41:40.031 -> temperature : 21.70 C
21:41:42.103 -> temperature : 20.50 C
21:43:05.338 -> temperature : 20.80 C
21:43:13.234 -> temperature : 23.20 C
```

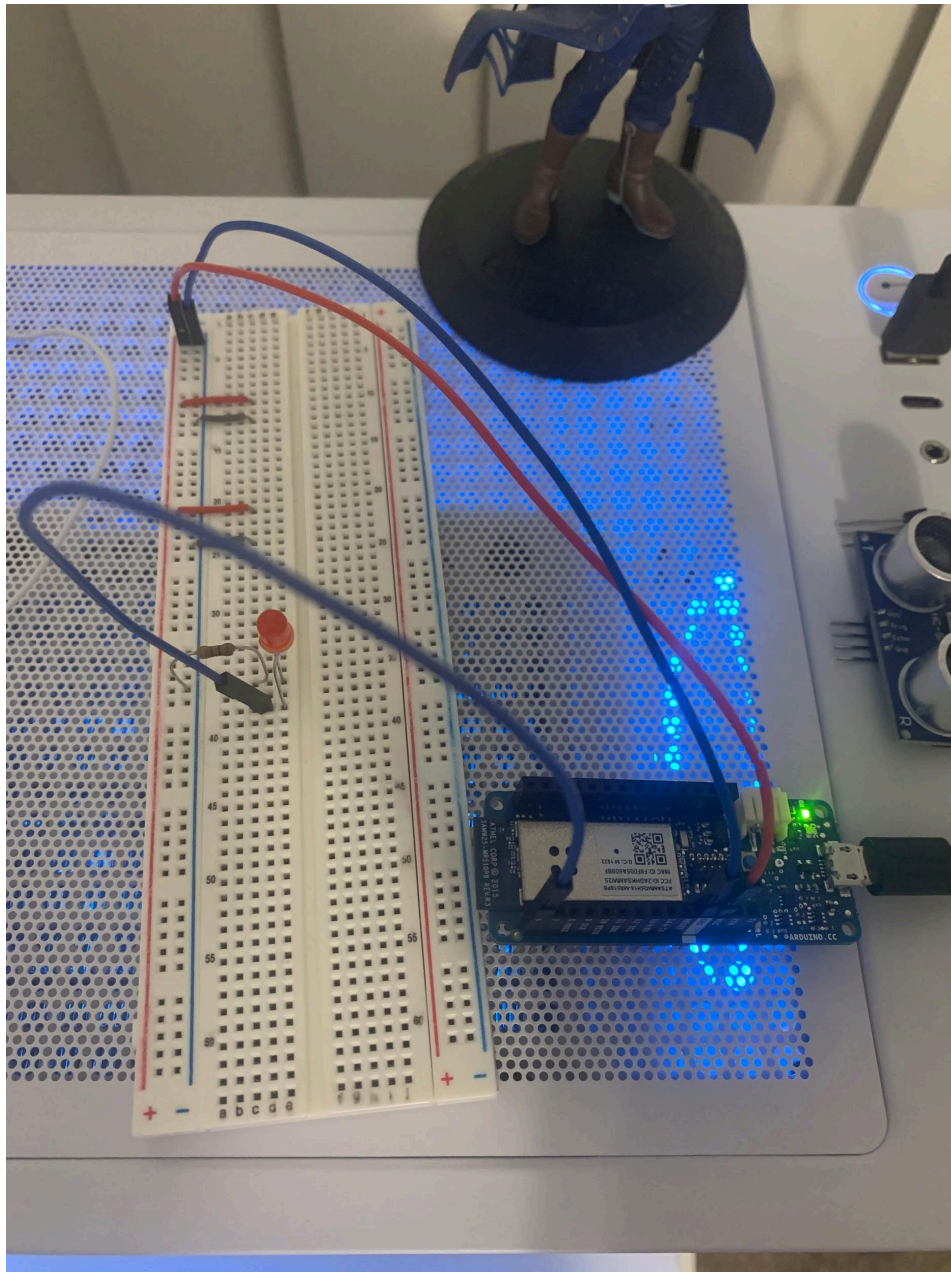
7.1 PHOTO SETUP:



7.2 YOUTUBE LINK:

<https://youtube.com/shorts/IMf11aN-8dc?si=K9GEGsUMUXRIcKGx0>

7.3 PHOTO SETUP:



7.4 YOUTUBE LINK:

<https://youtube.com/shorts/LLPURK9yo88?si=iL8ow2uyhoHlzi2t>

LED CODE:

```
// Template ID, Device Name and Auth Token are provided by the Blynk.Cloud
// See the Device Info tab, or Template settings
#define BLYNK_TEMPLATE_ID "TMPL2KIo5FwVI"
#define BLYNK_TEMPLATE_NAME "Kimberly"
#define BLYNK_AUTH_TOKEN "9hP9yGXO3NndpLtPTAjbZ_Ekuo-yn7BH"
int ledPin = 6; //led pin
// Comment this out to disable prints and save space
#define BLYNK_PRINT SerialUSB
#include <SPI.h>
#include <WiFi101.h>
#include <BlynkSimpleWiFiShield101.h>
// Your WiFi credentials.
// Set password to "" for open networks.
char ssid[] = "RojasFam";
char pass[] = "L0v3sh0t";
char auth[] = "9hP9yGXO3NndpLtPTAjbZ_Ekuo-yn7BH";

BLYNK_WRITE(V2)
{ // Executes when the value of v1 changes remember screen above
  if(param.asInt() == 1) // if true
  {
    sensor(); //execute sensor function
  }
  else{
    digitalWrite(ledPin, LOW); // turn led off
  }
}

void setup()
{
  // Debug console
  SerialUSB.begin(115200);
  Blynk.begin(auth, ssid, pass);
  // You can also specify server:
  pinMode(ledPin, OUTPUT); // set led pin assignment
  digitalWrite(ledPin, LOW); //automatically set led to off
}
```

```
void loop()
{
    Blynk.run();
}

void sensor()
{
    digitalWrite(ledPin, HIGH); // turn led on when button pressed
}
```